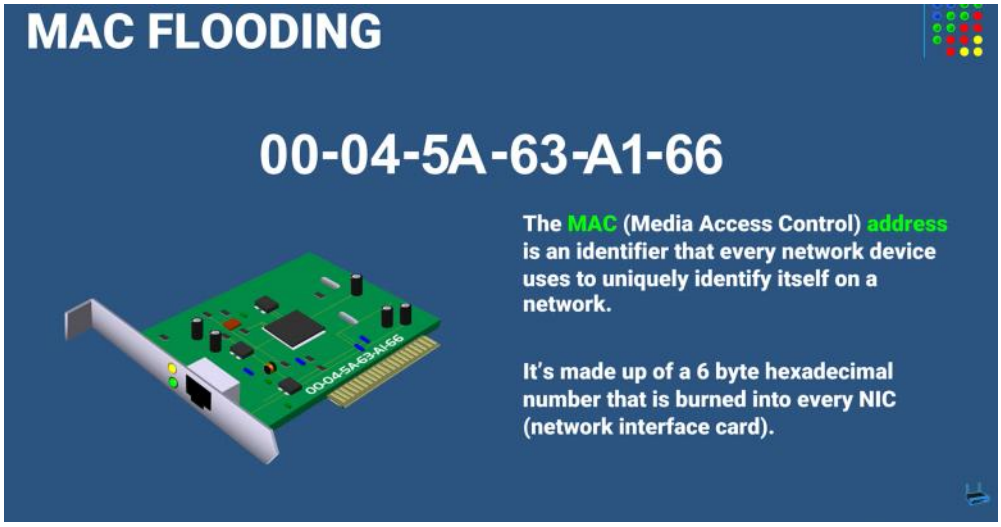
# MAC Flooding Attack

Sunday, April 12, 2026 11:46 AM

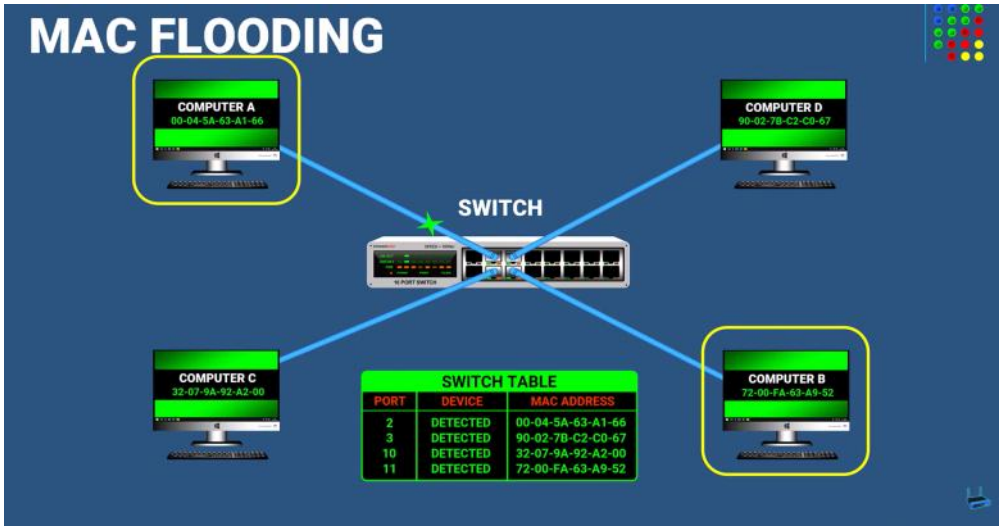
MAC Flooding is a network attack where a hacker overwhelms a switch with fraudulent MAC address entries. This is designed to disrupt network operations and expose sensitive data by forcing the switch into a "fail open" mode.

Key Concepts:

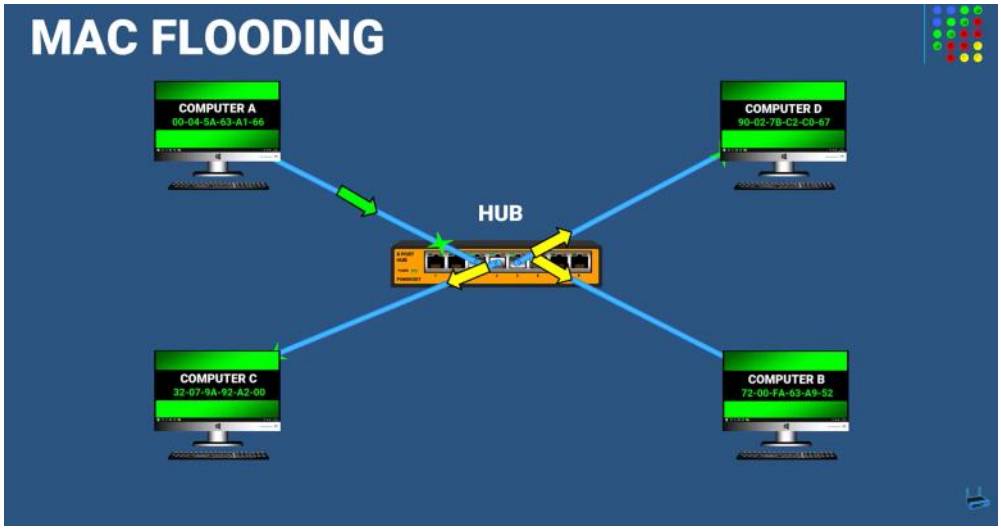
MAC Address: A unique, six-byte hexadecimal identifier assigned to every network interface card by the manufacturer to identify devices on a network.



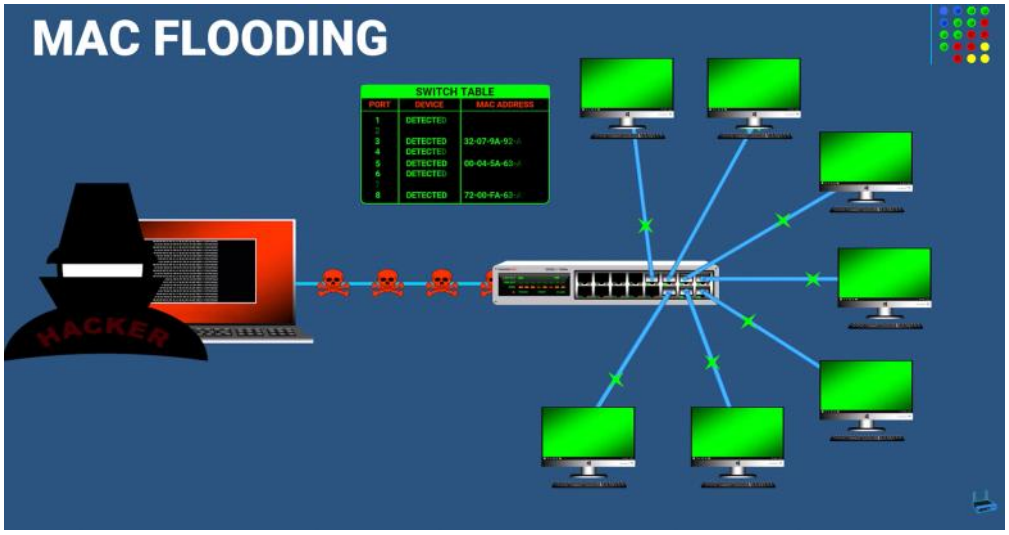

Switch: An intelligent networking device that connects devices and directs data frames only to the intended destination port by learning and storing MAC addresses in an internal table.




Hub: An older, non-intelligent networking device that does not use a table. It broadcasts incoming data frames to every port, which creates unnecessary traffic and poses significant security risks.




The Attack: By flooding a switch with fake MAC addresses, the device's memory is overwhelmed. The switch then defaults to "fail open" mode, acting like a hub and broadcasting traffic to every port, allowing an attacker to use a packet sniffer(a program) to capture and steal sensitive data.




Prevention Methods:

- Port Security: Limits the number of new MAC addresses a switch can learn, preventing the table from being overwhelmed.
- MAC Address Filtering: Configures the switch to accept data traffic only from authorized or legitimate MAC addresses.